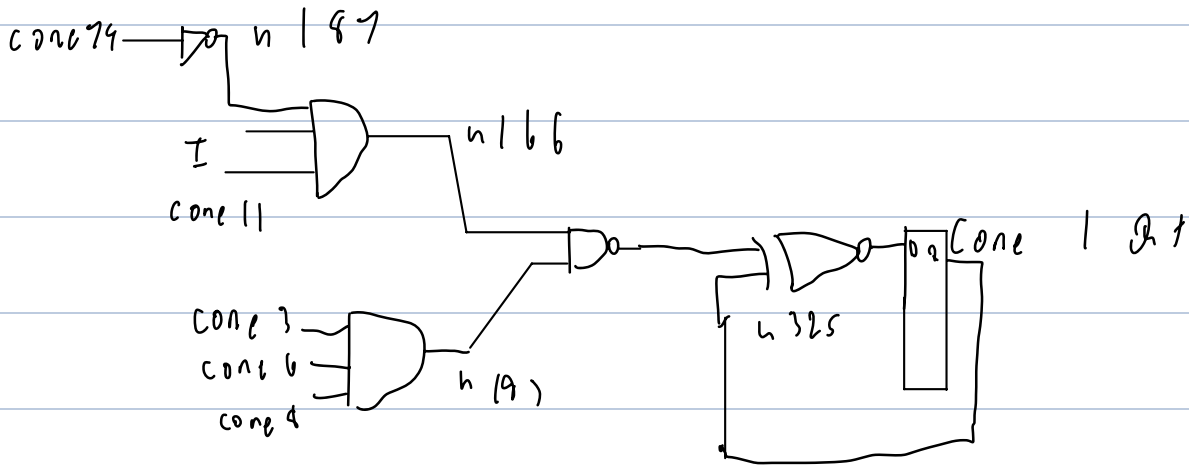


Conc analysis

first look at coner with inputs I

Conc 1 check



Behaviour conc 1 is zero until

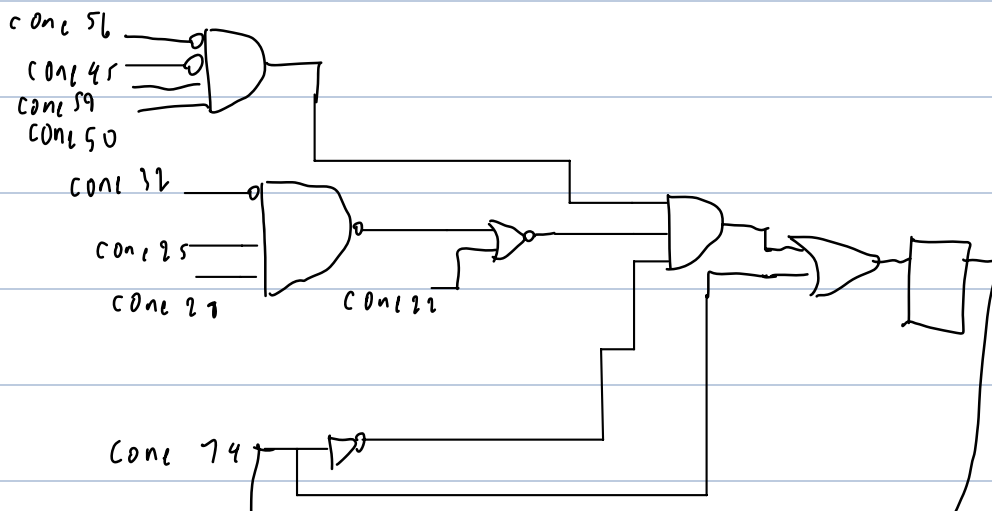
conc 74 is zero, I is 1, conc 11 is 1, conc 3, 6, 4 are

1. Then it sticks at 1 until the boolean

logic occurs again.

Conc 74.

This is also sticky but just is turned on the off completely. Also control. conc 74



cone be heavier and logic to point out.

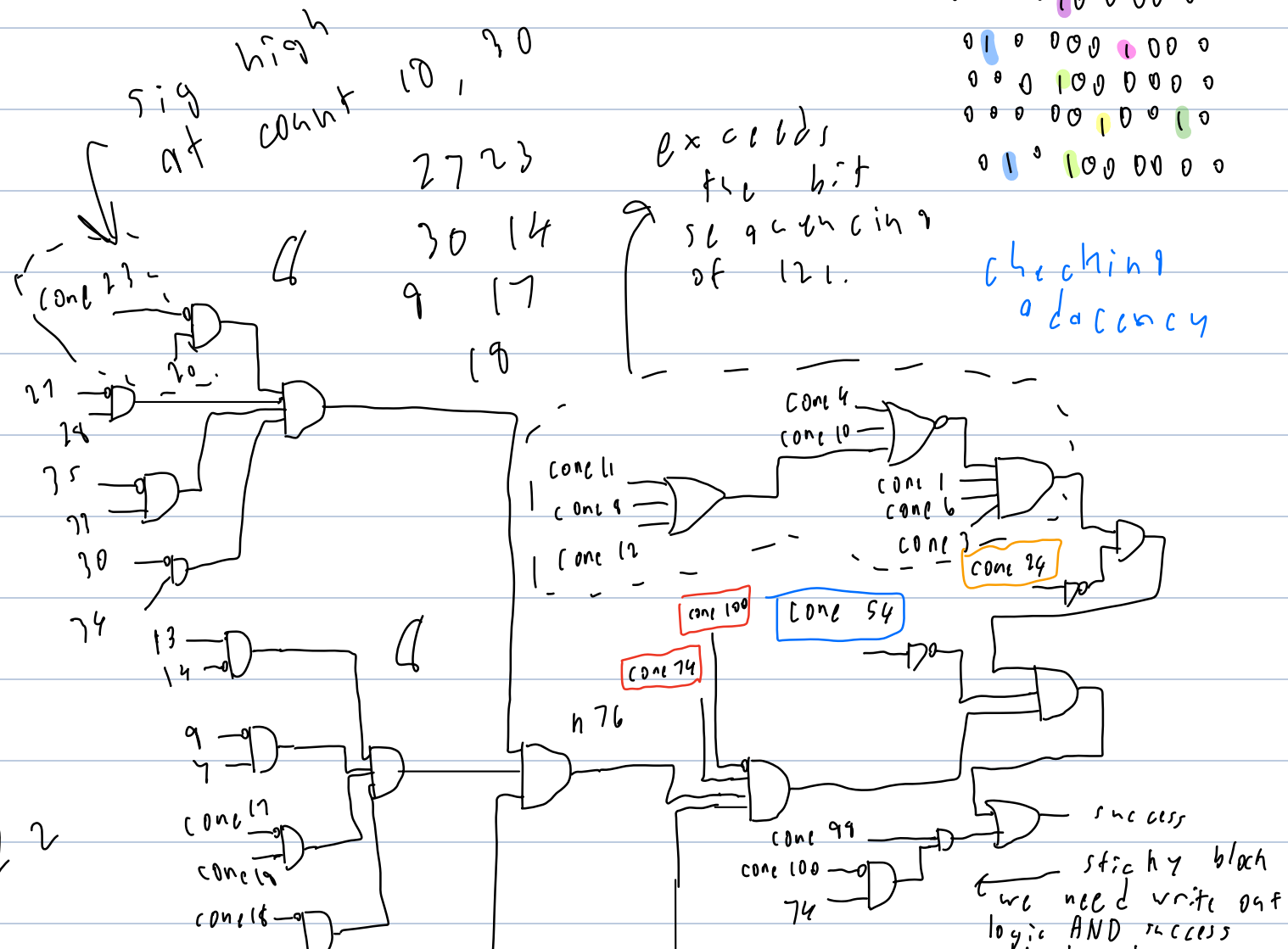
cone 74 had cones very close together making the logic. Investigate how that may have to do with what the logic is saying.

Wrong approach. Start from output and work back error?

Start with success cone 91
 Biggest block

```

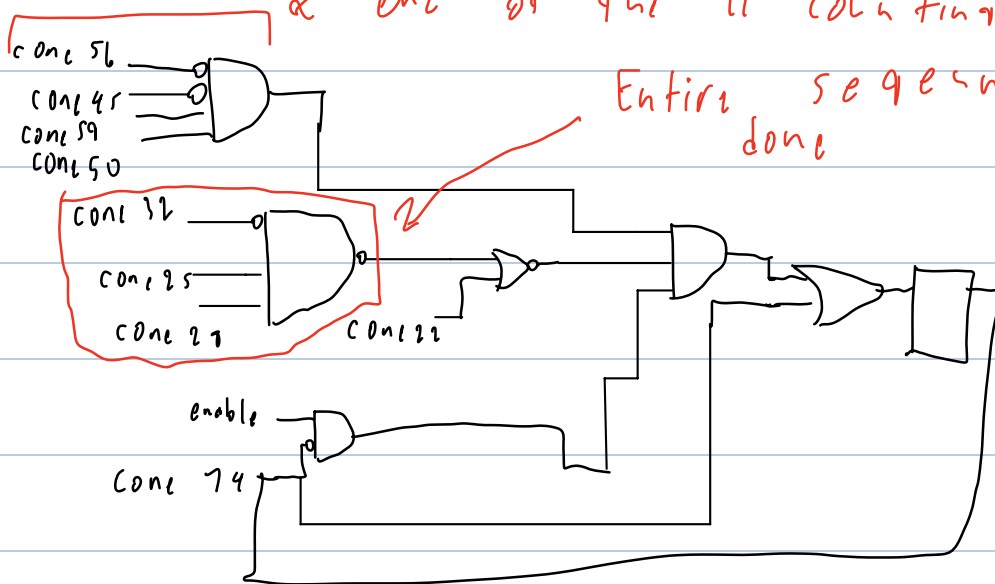
0 0 0 0 0 0 0 1 0 1 0
1 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 1 0 1 0
1 0 1 0 0 0 0 0 0 0 0
0 0 0 0 1 0 0 0 0 0 0
0 0 1 0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0 0 0 0
0 1 0 0 0 0 0 0 0 0 0
0 0 0 1 0 0 0 0 0 0 0
0 0 0 0 0 1 0 0 0 0 0
0 1 0 0 0 0 0 0 0 0 0
0 0 0 0 0 1 0 0 0 0 0
0 0 0 0 0 0 1 0 0 0 0
0 1 0 1 0 0 0 0 0 0 0
  
```



Let's under a four cone 74 first

at end of the 11 counting sequence.

Entire sequence is done



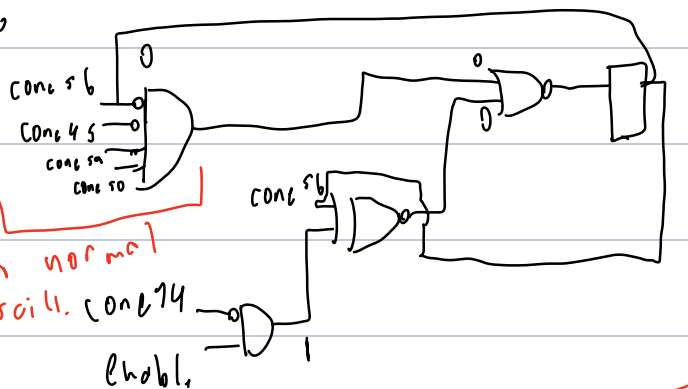
Cone 74 is also sticky. So both

cone 100 and cone 74 are holding 74 high once it goes high

Relies next on cone 22, cone 29, cone 25, cone 32,

cone 56, cone 45, 59, 50

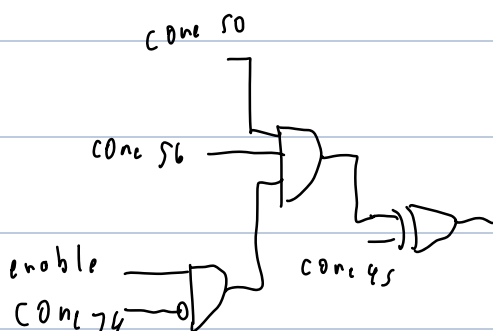
Cone 56

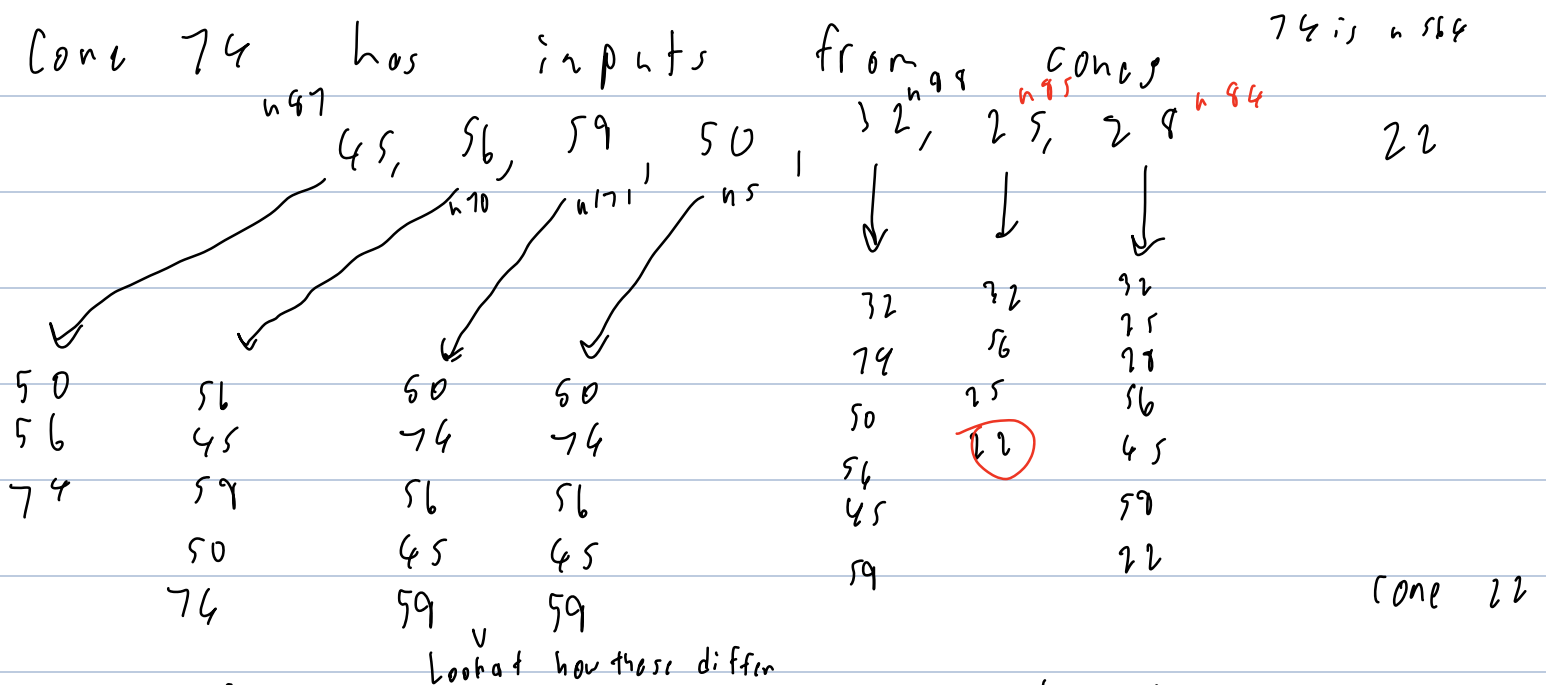


This is saying it will provide zero creating normal functioning oscill. cone 74 But under these terms.

We can say $\overline{56} \cdot \overline{45} \cdot 59 \cdot 50$ is the end of 11 term

Cone 45





Obviously control logic but all other inputs tell us more about control

Control logic is 9 concs

74, 22, 25, 29, 32, 45, 50, 56, 59

Analyze wave forms to give more information

key things to note this has no connection to input pins or data pins meaning it this is only based on clock cycles.

conc 22: This a 22 - sequence counter

conc 25: This a 66 - sequence counter

conc 29: This is a 44 - sequence c

conc 32: This is a 88 - sequence c

conc 74: This is trigger when count is 121

conc 56: A 2 sequence counter that passes

at count 11.

con. 59: Creates a high signal for pulses 8-11 for every repeating.

con. 45: Creates high signals between 4 and 9

con. 50: sequence 4 counter.

Analysis: So the state machine is in idle until enable.

After which con. 56 acts as a ² sequence counter

Next analysis is the two other inputs to the AND gate for success being con. 59 and a whole group

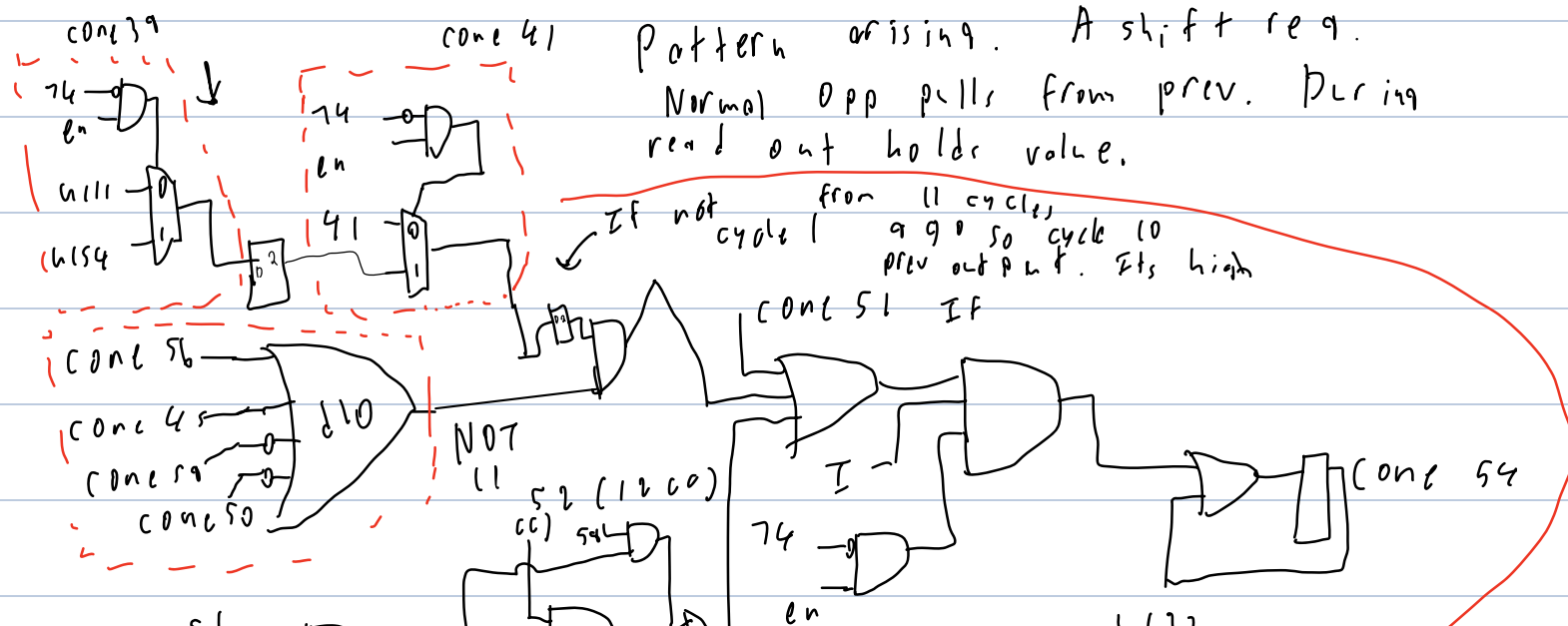
Con. 59.

I from 10 cycles ago

Pattern arising. A shift reg.

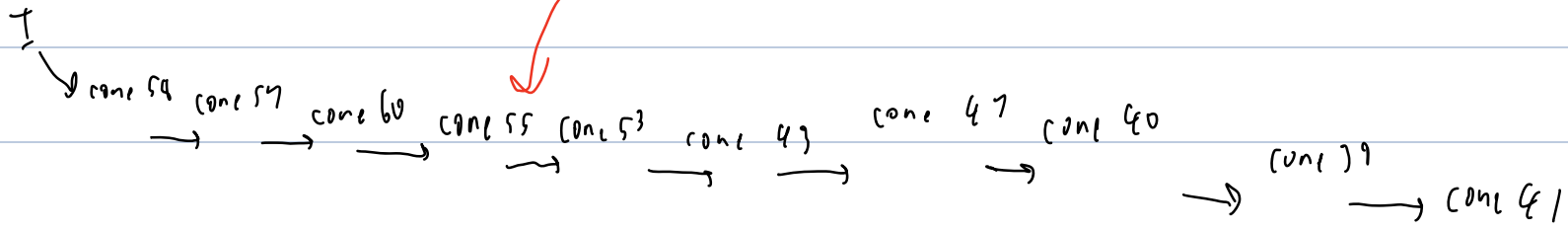
Normal opp pulls from prev. During read out holds value.

If not cycle 1 from 11 cycles, 9 9 0 50 cycle 10 prev output. It's high

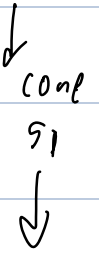




At any cycle except
to check if repeating
1 over
columns



conc 52 = Current theory is logic for repeating
signals



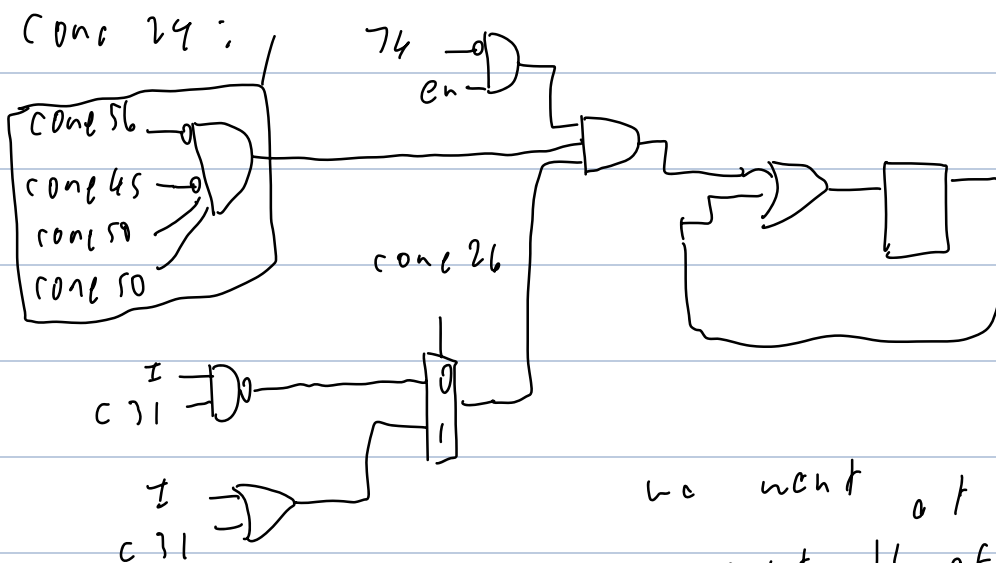
52: checker for signals that differ by 1 BUT BY column (1)

so if in word 1 there is 1 then there cannot be a
1 12 cycles before, 1 cycle before or 10 cycles before.

so in column now it needs control logic to stop

two flags 1 000000001 is a 10 cycle diff but

conc 11



conc 31 is a
high sig detector

conc 26
is a double
signal

detector

we want a

count of cycle the c31 to
be 200 and no 1 value if conc 26
already high but if not we want
both c31 to be
high and conc 26

conc 31

conc 31 is a 1 if conc 21 is triggered

signal that is

high from the first

1 in an 11 bit sequence

and then drops at

the second

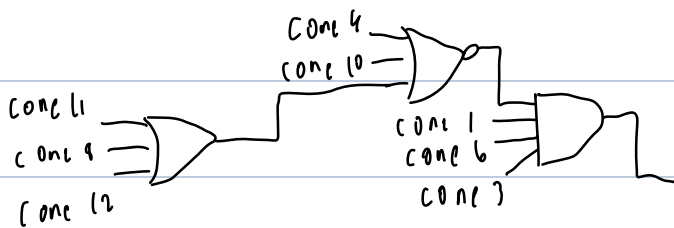
stay high

if conc 26 already high this is blocked

every thing but

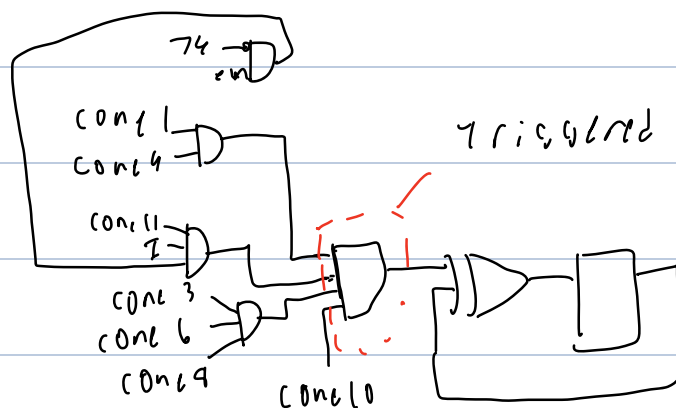
Conc 24, Needs to stay low and checks each 11 bit sequence only holds two values.

966 circuit



conc 9 is counting every 9 high signals

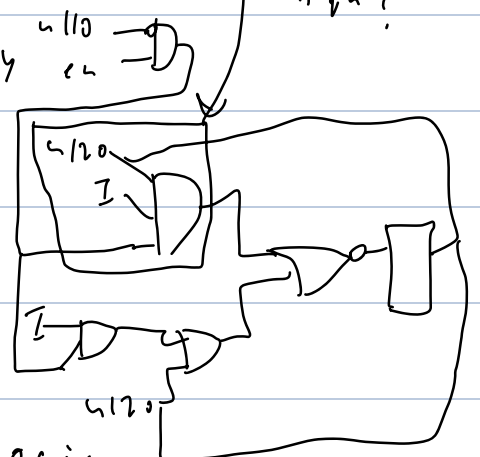
conc 12



Conc 11

Just a sticky signal detector alternates after every high signal

Is it currently high?



triggered logic

conc 1 counts lb high signals

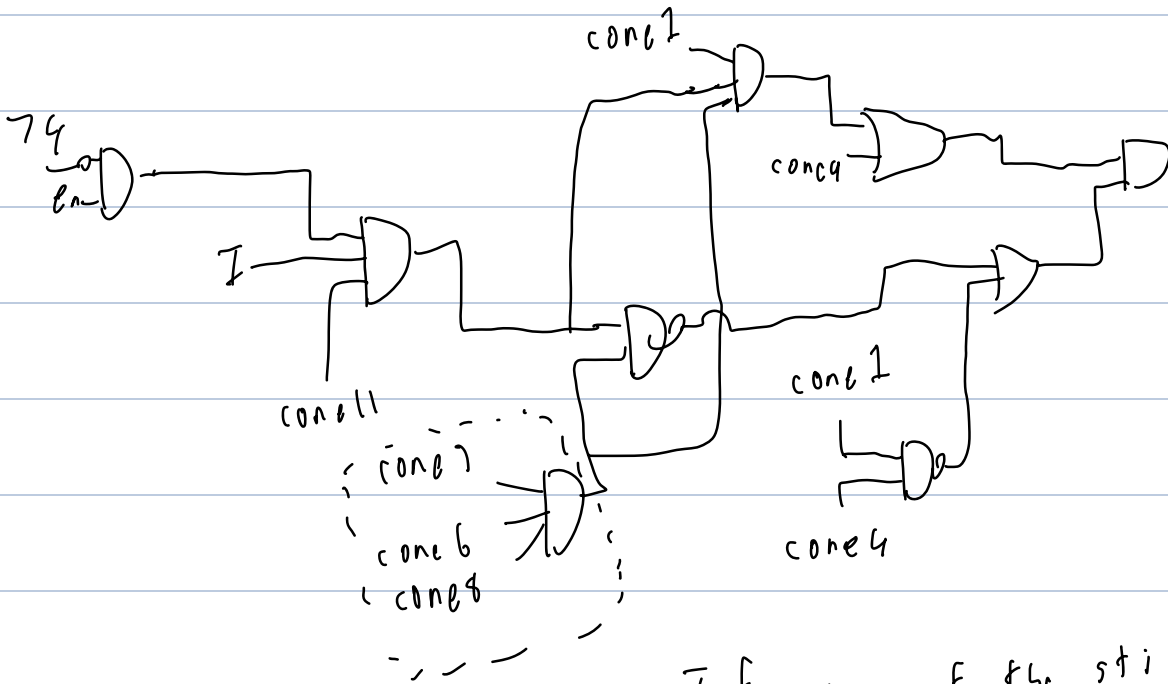
conc } counts every two high counts

come 4 further analysis

conc b counts ever 4.

come 4 is every 32 l's come 10 every 64

cone 9 counts every 32 cone 12 is every 128



If any of the sticky columns are high or cone 11 is high and/or 1 comes in

This is checking really if

New rule, has 1 on bit count.

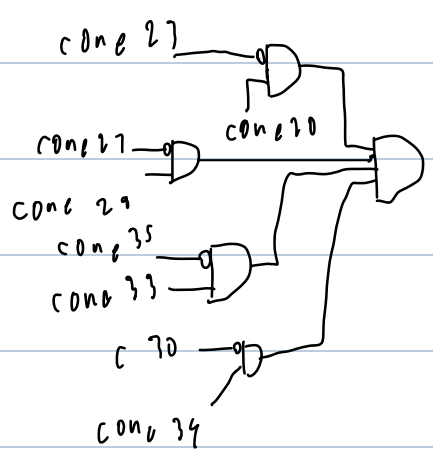
concl $1, 6, 7$ high $16 + 9 + 2 = 27$

con 4 8 10 11 12 low

22 ones in the solution

And every word has 2 ones based on conc 24

And the cones in one word mean the cones in the same next or prior cannot occur a neighbouring row diagonals included

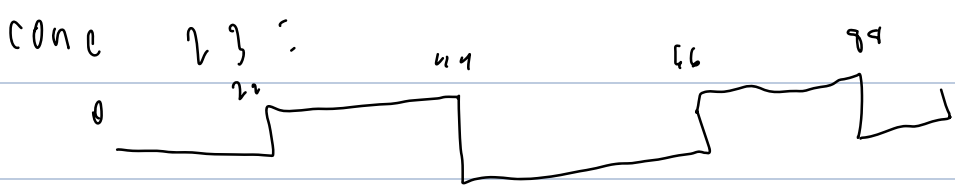


cone 20, 21, 27, 29, 70
73, 74, 75

cone 20, 23
inputs: cone 22, 45, 59, 52, 74, 32, 26

Zt control logic with pin 23

Look into later



cone 23 Zt analysis is

has to be 45, 56, 59, 50, 72, 25, 29, 22
 n47, n70, n71, n5, n99, n75, n84, n71
 sequence 0-11

22	29	32	45	50	56	25	59	74	22	23	20
0	-	0	0	0	1	0	1	0	1	1	0

needed to be on odd 11-bit sequence
 so this wants a specific number
 to turn zero.

At $I = 0$ at cycle 10 and some

output toggles if at cycle 10 I is high. This is falling as the of

hits

This is a check that in the final result there are exactly two ones. So first cone 23 goes high at the first 10. Then the next 10 turns cone 23 low because the toggle low is the exact same as toggle up. But now 23 is high if any more 10's occur the input sticks. Instead of flagging more 10's the logic is just expecting the same for all other inputs that this occurs.

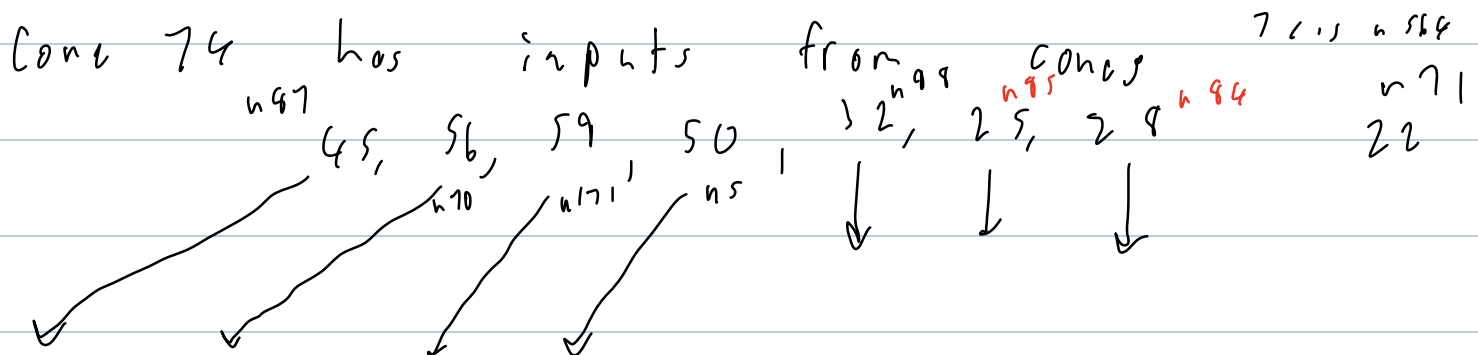
So

Next cone 27 29

Cone 27 to 1

0	1	0	1	0	0	0	0	0	1	1	0	0
74	en	22	24	45	56	59	I					
				32	50	25						29
												17

22 56946461 - 22
29 50940200 - 44



BIT list of cones: (count 80/99 90 MORE)

FSM Logic: 22, 25, 28, 32, 50, 56, 59, 74, 45, 100 (10)

Success driver: 99 (1)

17 stage shift reg: 39, 40, 41, 43, 47, 53, 55, 57, 60, 54, 51, 52 (12)

Conc 52: Adjacency checks (1)

Bit counting logic (per count): 24, 26, 31 (7)

Bit count (global): 11, 12, 8, 1, 7, 6, 10, 4, 12 (9)

Bit Checking logic: + 22 gates, + 22

Out buff: 8 + 95 + 57 + 62 + 63 + 65 + 68 + 69 + 70 + 71 + 83 + 89

TO DO:

Write Up

22 lower side of success

Out buff logic

Conc 63: 24, 28, 43, 71, 68, 63, Conc 100
62

Conc 67

Some sort of control?

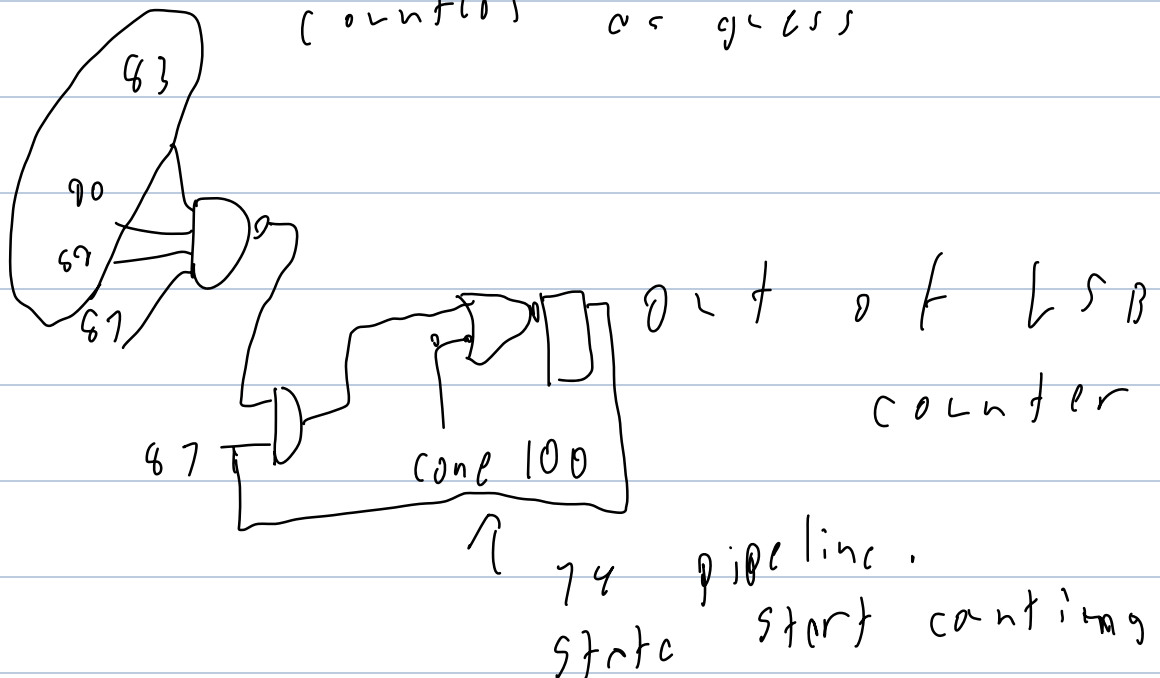
pairs all the same and inputs to drive
/ are ambiguous

0 [0] cone 73
in waves shows

25 must be high
and 26 d l

0	1	0	-	0	0	1	0	1	0	1	0
74	en	22	27	32	45	50	56	25	59	75	77

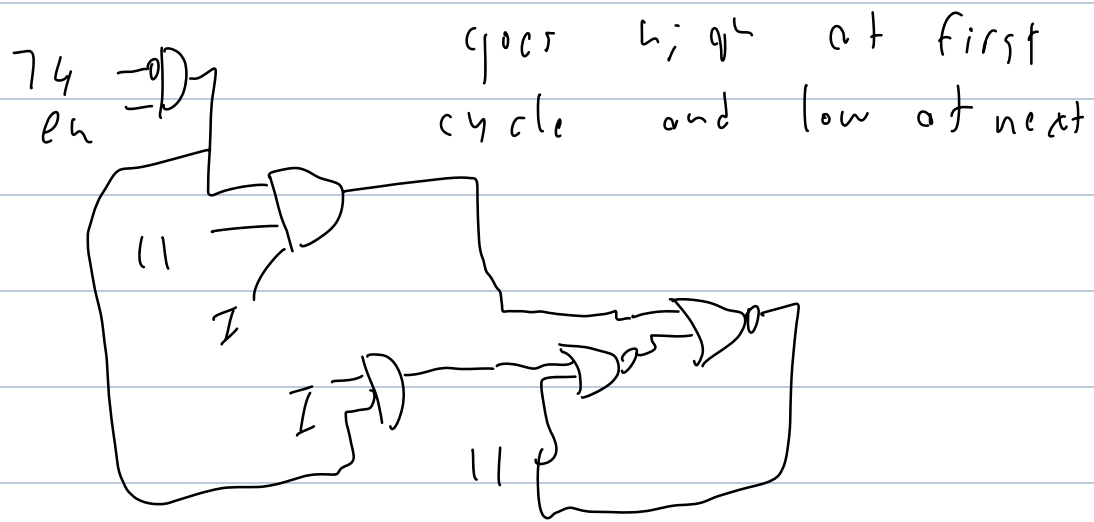
cone 87 counters as guess



counter for output reg

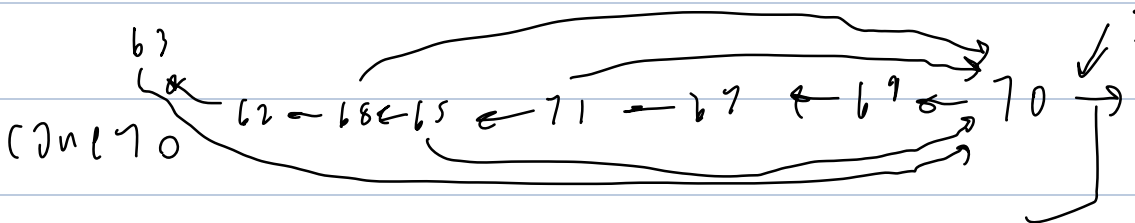
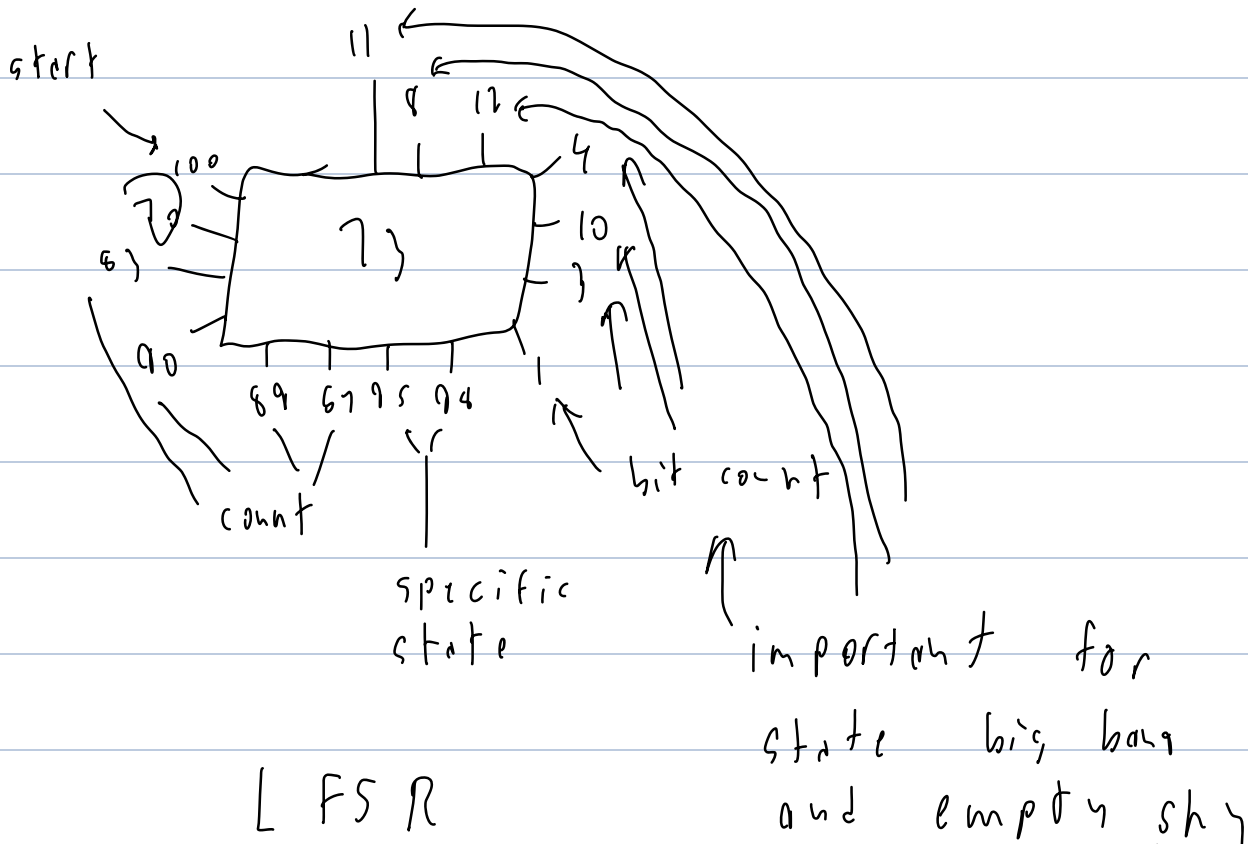
83, 90, 89, 87

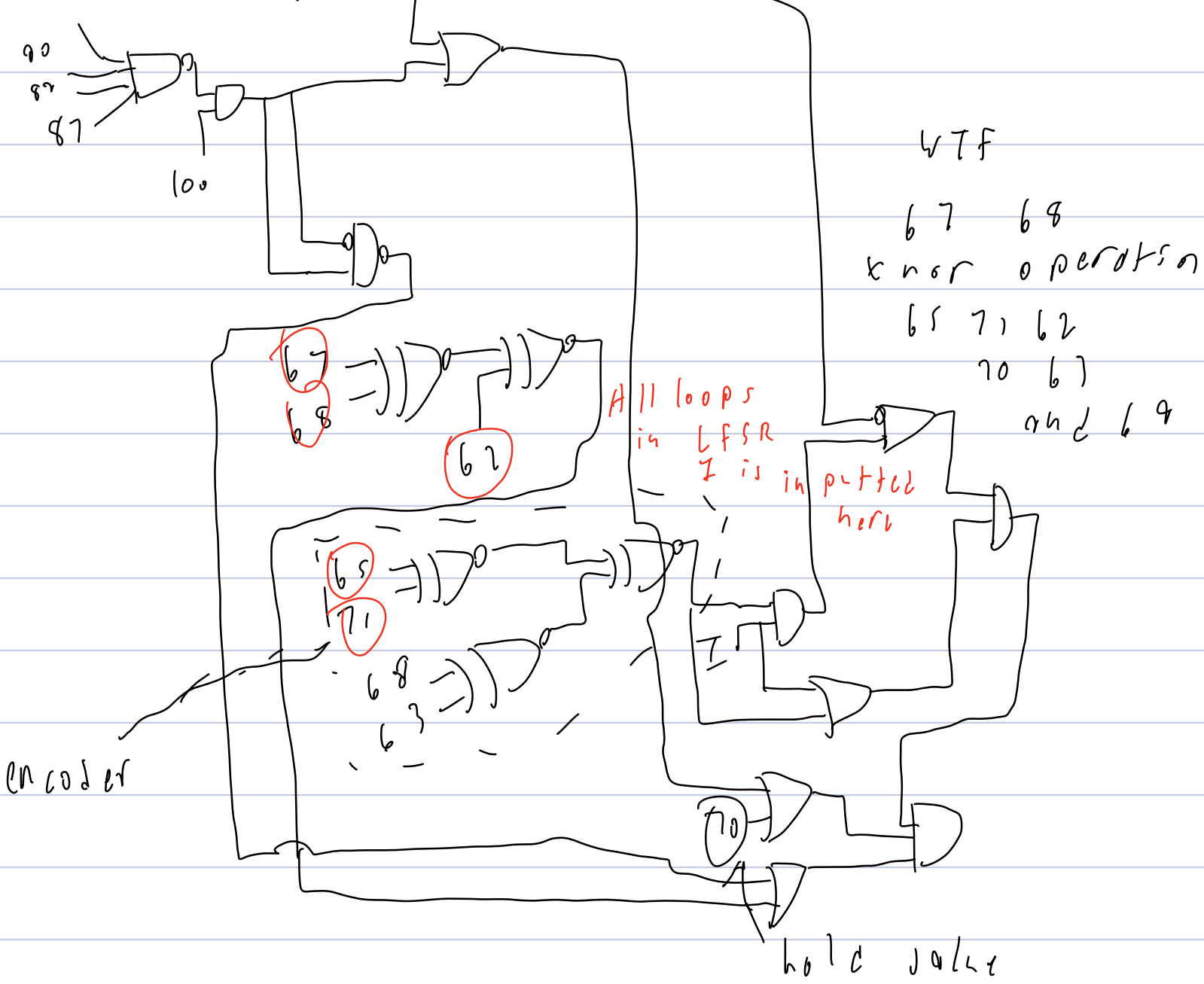
cong 11



cong 4

Ph want decoding it counting 22 bits
Inputs to output block





what actually triggers each cone

conv 77

8-11

4 main time frames

74 en 22 32 50 59
 0 | 0 - 0 | 1 1 0 0 1 0 -

between 8 at 0-11 and 22-33

0 | 1 0 0 0 0 1 1 0 - between 9 at 22-44

30	30	30	30	30	42	42	22	27	23	27
30	30	9	30	30	42	22	22	27	23	27
30	30	9	42	42	42	42	22	22	23	27
30	30	9	42	14	14	14	22	22	23	27
9	30	9	42	14	14	14	22	22	23	27
9	9	9	42	14	14	14	22	22	23	27
42	42	42	42	42	42	14	22	22	23	27
42	35	35	35	14	14	14	22	22	23	27
42	35	35	17	35	35	35	22	22	23	27
42	35	35	17	35	35	35	22	22	23	27

56 50

0 | 1 | 0 0 | 1 - 0 0 1 0 - between 22 - 33 7-8

0 1 1 | 0 0 0 - 0 1 1 0 - between 33 - 44 10

cone 22: This all - sequence counter

cone 25: This a 64 - sequence counter

cone 29: This is all - sequence c

cone 31: This is a 44 - sequence c

cone 24: This is trigger when count is 121

cone 56: A 2 sequence counter that pauses

cone 2⁹ doesn't help all found at count 11.

cone 59: Creates a high signal for pulses for every repeating.

cone 45: Creates high signals between 4 and 8

cone 50: sequence 4 counter.

cone 30

27 ~ 3

30 14

9 17

19

74 en 22 28 50 32 45 56 25 59 1 30 34 1, 2 cyc 0-44

0 1 - - 0 0 0 - 0 0 1 0 -

0 1 - - 0 0 1 0 0 1 1 0 - Not pos

0 1 0 0 - 0 1 0 0 1 1 0 - X

0 1 0 0 1 0 - 1 0 1 1 0 - X

0 1 0 0 0 - 0 1 0 0 1 0 - 11 - 22 cyc 1, 5

0 1 - 0 0 0 - 0 0 0 1 0 - 0 - 22 cyc 4

0 1 0 0 - 0 0 - 0 0 1 0 - 0 - 11 cyc (2) 4

0 1 0 - 1 0 0 1 0 1 1 0 - X

cone 2 }
2228 45 50 56 25 59 74 en I 2320

- 000 0 - 0 1 0 1 0 - 0 - 22 cyc 9,10
0 - 00 0 1 0 1 0 1 1 0 0 - 11 22-33 cyc 10

cone 14

2228 72 45 50 56 25 59 74 en I 141 }
1 1 - 1 - 0 0 0 0 1 1 0 - 44-55 66-77 cyc 77
- 0 1 1 0 0 0 0 0 1 1 0 - 44-55 cyc 45
1 - 1 1 0 - 0 0 0 1 1 0 - 44-66 cyc 57
1 - 1 1 - 0 0 0 0 1 1 0 - 44-55 cyc 57 66-77
- 1 1 1 1 0 0 0 0 1 1 0 - 66-77 cyc 67
1 1 - 1 0 - 0 0 0 1 1 0 - 44-66 cyc 37
- 1 0 1 1 1 0 1 0 1 1 0 - x

cone 9

cone 3 }
cyc 2
45-110
cyc 1 44-99
110-121
77-89 {
77-89 2 }
22-33 cyc 2,3
none
none
22-33 C 1 }
1V 0
1V 0
44-66 C 1 }
55-66 C 0 }

cone 18
66-88 C 8
88-11 C 8
55-66 89
99-110 811
44-10 84

concl 39 1

concl 17

42 44

concl 42